

PBA Research Project Final Report Evaluation - Group 3

Project Title: Policy Strictness and Heterogeneous Mobility Responses: A Natural Experiment on Workplace and Park Mobility in Denmark and Sweden

Group #: 3

Authors: names and student IDs removed

I. Research Question & Objectives (30% weight)

Clarity and Specificity of Research Question(s)

Score: 4.5/5

The research question leverages a compelling natural experiment: Denmark and Sweden's divergent COVID-19 policy responses provide quasi-random variation in policy strictness between otherwise similar Scandinavian countries. The focus on heterogeneous mobility responses across employment and poverty dimensions adds analytical depth beyond a simple aggregate comparison.

Feasibility of Objectives

Score: 4.5/5

The objectives are well-scoped and achievable. The choice to focus on workplace and park mobility as outcomes is sensible given data availability (Google Mobility Reports). The heterogeneity analysis by employment sector and poverty rates is ambitious but feasible.

II. Methodology (25% weight)

Appropriateness of Methods

Score: 5/5

The difference-in-differences design is well-suited to this setting:

- **Treatment:** Denmark's stricter lockdown policies
- **Control:** Sweden's lighter-touch approach
- **Timing:** Clear policy divergence in March 2020

Critically, this is a **valid DiD design** because the treatment (Denmark's policy choice) was determined before and independently of the outcome window, likely generating exogenous variations that can be examined. This is fundamentally different from designs where treatment is defined by realized behavior during the study period.

The inclusion of date fixed effects appropriately controls for common time shocks. The heterogeneity analysis examining interactions with employment composition and poverty rates demonstrates rigorous thinking about effect modification.

Rigour and Ethical Considerations

Score: 4.5/5

The analysis is rigorous with appropriate attention to identification assumptions. The parallel trends assumption is discussed, though more formal testing (e.g., examining pre-treatment mobility trends)

would strengthen the case. The acknowledgment that Denmark and Sweden may differ in unobserved ways affecting mobility is appropriately cautious.

The finding that poverty interacts with park mobility in a direction opposite to the initial hypothesis, and the mature handling of this "counter-hypothetical" result, demonstrates intellectual honesty and careful interpretation.

III. Results/Findings (15% weight)

Clarity and Presentation

Score: 4.5/5

Results are presented clearly with well-organized tables and figures. The progression from main effects to heterogeneity analysis is logical. Visualizations effectively communicate the mobility divergence between countries.

Interpretation and Significance

Score: 4.5/5

The interpretation is careful and nuanced. The discussion of why poverty might have unexpected effects on park mobility (e.g., urban density, access to private outdoor space) shows thoughtful engagement with surprising findings rather than dismissing them or forcing them into the original hypothesis.

IV. Discussion & Conclusion (15% weight)

Relation to Existing Literature

Score: 4/5

The literature review situates the work within the growing body of COVID-19 policy evaluation research. The connection to behavioral economics and compliance literature is appropriate. Integration with epidemiological findings on mobility and transmission would add depth.

Conclusions and Implications

Score: 4.5/5

The conclusions are appropriately scoped. The discussion of policy implications—that one-size-fits-all mobility restrictions may have heterogeneous effects across demographic groups—is policy-relevant and well-supported by the analysis.

V. Writing Quality (15% weight)

Organization and Structure

Score: 4.5/5

The report is well-organized with clear sections. The natural experiment framing is established early and maintained throughout. The heterogeneity analysis is integrated smoothly rather than feeling tacked on.

Clarity and Precision

Score: 4.5/5

Writing is clear. Causal language is appropriately disciplined given the natural experiment design, the DiD framework with a clear policy shock provides stronger grounds for causal interpretation than many observational studies.

General Comments

This is a well-executed project that demonstrates outstanding engagement with course concepts and iterative refinement through multiple consultations. Your active participation in discussions throughout the semester, taking careful notes and revising your approach based on feedback, is clearly reflected in the quality of the final product. Among all groups, you engaged most frequently and substantively with the material. The heterogeneity analysis by employment and poverty adds analytical value.

The identification strategy here is sound: Denmark's policy choice was made before the outcome period and provides genuine exogenous variation. This is the correct way to implement DiD.

Areas for Further Development

- Consider more formal parallel trends testing using pre-COVID mobility data
 - Discuss potential threats to identification beyond parallel trends (e.g., differential testing rates, reporting differences)
 - The poverty interaction finding deserves even more exploration, this could be a paper in itself
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Class Performance Context

Statistic	Value
Class Mean	84
Class Median	82
Standard Deviation	5.3
Range	77 – 92
Your Score	90

Overall Score: 90 / 100

(This will contribute 22.5 / 25 to your final letter grade)

Score Breakdown

Section	Weight	Score
I. Research Question	30%	4.5/5, 4.5/5
II. Methodology	25%	5/5, 4.5/5
III. Results	15%	4.5/5
IV. Discussion	15%	4.25/5
V. Writing	15%	4.5/5
TOTAL	100%	90/100